

PROJECT DESIGN PHASE III

PROPOSED SOLUTION FIT

Date	02 October 2026
Team ID	SWTID-2026-1180
Project Name	Implement Client Script and UI Policy (incidents)
Maximum Marks	2 Marks

PROPOSED SOLUTION

Proposed Solution: Client-Side Orchestration & Form Optimization

To resolve form clutter, eliminate server-side validation delay friction, and ensure pristine data integrity, the project proposes a robust **client-side orchestration architecture** built directly inside ServiceNow.

The solution replaces slow, full-page reloads with instant, browser-level controls using the following core components:

1. Dynamic Form Orchestration (No-Code UI Policies)

- **Conditional Visibility:** Automatically shows or hides specific input fields based on user selections (e.g., displaying asset tags and serial numbers only when the category is set to `Hardware`).
- **Contextual Enforcement:** Dynamically switches fields between mandatory, optional, or read-only states in real-time, eliminating cognitive overload and reducing visual clutter by up to 60%.

2. Form Initialization & User Context (`onLoad` Client Scripts)

- **Automated Profile Setup:** Automatically queries session attributes (`g_user`) upon form load to pre-populate caller name, email, department, and contact preferences.
- **Streamlined Entry:** Removes redundant manual data entry for standard users and support agents right from the start.

3. Real-Time Data Validation (`onChange` Client Scripts & Regex)

- **Live Format Checking:** Evaluates syntax regular expressions (Regex) instantly as the user types phone numbers, email addresses, or specific date codes.
- **Inline Helper Messaging:** Instantly displays non-blocking helper warnings or errors directly beneath the target field using `g_form.showFieldMsg()`, allowing immediate correction before submission.

4. Non-Blocking Asynchronous Lookups (`GlideAjax` & Script Includes)

- **Background Data Verification:** Offloads heavy database queries—such as checking Configuration Item (CI) operational status, maintenance contracts, or warranty coverage—to server-side Script Includes.
- **Zero UI Lag:** Communicates asynchronously in the background via `XMLHttpRequest` so the user interface never freezes during data lookups.

5. Pre-Submission Gatekeeper (`onSubmit` Client Scripts)

- **Final Validation Check:** Acts as the last line of defense, scanning all required inputs and validation flags right when the user clicks **Submit**.
- **Safe Interception:** If any errors or malformed data are detected, submission is safely aborted (`return false`) directly in the browser. This completely prevents full-page reloads and protects all typed notes from being lost.